CKAT EXAM PREPARATION

Time & Distance

Complete Study Guide

Formulas · Conversions · Units · Concepts

13 Graded Practice Questions · Shortcuts & Tricks

■ **Section A – Quantitative Aptitude**

Up to 12 Questions in Exam | Topic 1 of 26

SECTION 1 — ALL FORMULAS

Core Relationship

Distance (D)	=	Speed (S) × Time (T)
Speed (S)	=	Distance (D) ÷ Time (T)
Time (T)	=	Distance (D) ÷ Speed (S)

Speed Conversion Formulas

km/h → m/s	Multiply by 5/18	x km/h = x × 5/18 m/s
m/s → km/h	Multiply by 18/5	x m/s = x × 18/5 km/h
km/h → mph	Multiply by 0.6214	x km/h ≈ x × 0.6214 mph
mph → km/h	Multiply by 1.6093	x mph ≈ x × 1.6093 km/h

Average Speed Formulas

Case	Formula
Equal distances, different speeds S1 and S2	**Avg Speed = 2S1·S2 / (S1+S2) [Harmonic Mean]**
Equal distances, 3 speeds S1, S2, S3	**Avg Speed = 3S1·S2·S3 / (S1S2+S2S3+S3S1)**
Equal time, different speeds S1 and S2	**Avg Speed = (S1+S2) / 2 [Arithmetic Mean]**

Relative Speed & Special Formulas

Relative speed (same direction)	**S_rel = |S1 – S2|**
Relative speed (opposite directions)	**S_rel = S1 + S2**
Time to cross a stationary object	**T = Length of train / Speed of train**
Time to cross a platform/bridge	**T = (Length of train + Length of platform) / Speed**

Time to cross a moving object (opp.)	**T = (L1 + L2) / (S1 + S2)**
Time to cross a moving object (same)	**T = (L1 + L2) / |S1 – S2|**

Meeting & Circular Track Formulas

Two people walk towards each other	**Time to meet = D / (S1+S2)**
Two people walk in same direction	**Time to meet = D / |S1–S2|**
Circular track — opposite dir.	**Time to meet = Track length / (S1+S2)**
Circular track — same direction	**Time to meet = Track length / |S1–S2|**
A meets B, reaches B's start in t1, A reaches A's start in t2	**S1/S2 = sqrt(t2/t1)**

SECTION 2 — CONVERSIONS & UNITS

Speed Conversions

Unit Pair	Factor	Example
km/h → m/s	× 5/18	72 km/h = 72×5/18 = 20 m/s
m/s → km/h	× 18/5	15 m/s = 15×18/5 = 54 km/h
km/h → mph	× 0.6214	100 km/h ≈ 62.14 mph
mph → km/h	× 1.6093	60 mph ≈ 96.56 km/h
m/s → cm/s	× 100	5 m/s = 500 cm/s
cm/s → m/s	÷ 100	500 cm/s = 5 m/s
knots → km/h	× 1.852	10 knots = 18.52 km/h

Distance Conversions

1 km	=	1000 m	|	**1 m**	=	100 cm
1 km	=	0.6214 miles	|	**1 mile**	=	1.6093 km
1 mile	=	1760 yards	|	**1 yard**	=	3 feet
1 foot	=	12 inches	|	**1 inch**	=	2.54 cm
1 nautical mile	=	1.852 km	|	**1 furlong**	=	201.168 m

Time Conversions

1 hour	= 60 minutes	= 3600 seconds
1 minute	= 60 seconds	= 1/60 hour
1 day	= 24 hours	= 1440 minutes = 86400 seconds
1 week	= 7 days	= 168 hours

Standard Units Reference

Quantity	SI Unit	Common Units Used in Exams

Distance	Metre (m)	km, m, cm, miles, yards
Speed	m/s	km/h, m/s, mph, knots
Time	Second (s)	hours (h), minutes (min), seconds (s)
Acceleration	m/s^2	Rarely tested at this level

SECTION 3 — FUNDAMENTAL CONCEPTS

What is Speed?

Speed is the rate at which distance is covered. It tells us how fast an object moves. Speed is a scalar quantity (no direction). Average speed = Total Distance / Total Time.

What is Distance?

Distance is the total path length covered by a moving object. It is always positive. Distance = Speed × Time. When speed or time changes, calculate each segment separately and add.

What is Time?

Time is the duration taken to cover a certain distance. Time = Distance / Speed. If two trains start simultaneously from opposite ends, calculate time using combined speed.

Uniform vs Non-Uniform Speed

Uniform speed: equal distances covered in equal time intervals. Non-uniform speed: distance covered varies. In aptitude problems, unless stated otherwise, assume uniform (constant) speed.

Average Speed — Critical Concept

Average Speed ≠ Average of Speeds! For equal distances: Avg Speed = 2ab/(a+b). For equal time: Avg Speed = (a+b)/2. This distinction is a common exam trap.

Relative Speed

When two objects move, we consider speed relative to each other. Same direction: subtract speeds. Opposite direction: add speeds. Used heavily in train and boat problems.

Meeting Point Problems

When A and B walk toward each other, they meet after time T = D/(Sa+Sb). When walking away, they separate at rate Sa+Sb. When chasing, time = gap/relative speed.

Effect of Speed Change

If speed increases by x%, time decreases. If speed is n/m of original, time becomes m/n of original. Distance is constant in these problems.

Quick Time Conversion Reference

Fraction of Hour	Minutes	Seconds
1/4 hour	15 min	900 sec
1/3 hour	20 min	1200 sec
1/2 hour	30 min	1800 sec
2/3 hour	40 min	2400 sec
3/4 hour	45 min	2700 sec
1 hour	60 min	3600 sec
1.5 hours	90 min	5400 sec
2 hours	120 min	7200 sec

SECTION 4 — 13 PRACTICE QUESTIONS (Increasing Difficulty)

Q1 EASY

A car travels 150 km in 3 hours. What is its speed in km/h?

Solution: Speed = Distance / Time = 150 / 3 = **50 km/h**

■ *Direct application of S = D/T.*

Q2 EASY

Convert 72 km/h into m/s.

Solution: 72 × 5/18 = **20 m/s**

■ *Always use ×5/18 for km/h → m/s.*

Q3 EASY

A person walks at 6 km/h. How far will he walk in 2 hours 30 minutes?

Solution: Time = 2.5 h. Distance = 6 × 2.5 = **15 km**

■ *Convert 30 min = 0.5 h, then use D = S×T.*

Q4 MODERATE

A train 200 m long passes a pole in 10 seconds. Find its speed in km/h.

Solution: Speed = 200/10 = 20 m/s = 20 × 18/5 = **72 km/h**

■ *Passing a pole = train travels its own length.*

Q5 MODERATE

A and B start from the same point in opposite directions at speeds of 4 km/h and 6 km/h. After how much time will they be 20 km apart?

Solution: Relative speed = 4 + 6 = 10 km/h. Time = 20/10 = **2 hours**

■ *Opposite directions → add speeds.*

Q6 MODERATE

A man covers a distance at 60 km/h and returns at 40 km/h. Find his average speed for the whole journey.

Solution: Avg speed = 2×60×40/(60+40) = 4800/100 = **48 km/h**

■ *Equal distances, different speeds → Harmonic Mean formula.*

Q7 MODERATE

Two trains 150 m and 200 m long run at 45 km/h and 36 km/h in the same direction. In how many seconds will the faster train cross the slower train?

Solution: Relative speed = 45–36 = 9 km/h = 9×5/18 = 2.5 m/s. Total length = 350 m. Time = 350/2.5 = **140 seconds**

■ *Same direction → subtract speeds. Cross = sum of lengths.*

Q8 HARD

Without stoppages, a train travels 75 km/h; with stoppages, it covers only 60 km/h. For how many minutes does it stop per hour?

Solution: In 1 hour at 75 km/h, it would cover 75 km. But it covers only 60 km. Distance lost = 15 km. Time for 15 km at 75 km/h = 15/75 h = 1/5 h = **12 minutes**

■ *Think of lost distance due to stoppages.*

Q9 HARD

A thief is spotted by a policeman at a distance of 100 m. The thief starts running and the policeman chases him. If the thief runs at 8 km/h and the policeman runs at 10 km/h, how far does the thief run before being caught?

Solution: Relative speed = 10–8 = 2 km/h. Time to close 100 m = 0.1 km: T = 0.1/2 = 1/20 h. Distance run by thief = 8 × 1/20 = **0.4 km = 400 m**

■ *Find time from relative speed, then distance = thief's speed × time.*

Q10 HARD

A and B start from P and Q respectively (300 km apart) at the same time toward each other. A's speed is 50 km/h and B's is 70 km/h. After meeting, A takes t■ hours to reach Q and B takes t■ hours to reach P. Find t■.

Solution: Meeting time = 300/(50+70) = 2.5 h. At meeting: A covered 125 km, B covered 175 km. t■ = 175/50 = 3.5 h. t■ = 125/70 ≈ **1.786 hours (1 h 47 min)**

■ *After meeting, each covers what the other had covered.*

Q11 VERY HARD

Walking at 3/4 of his usual speed, a man is 20 minutes late. Find his usual time for the journey.

Solution: New speed = 3S/4. New time = 4T/3. Extra time = 4T/3 – T = T/3 = 20 min. Therefore T = **60 minutes**

■ *If speed = (n/m) × usual, then time = (m/n) × usual.*

Q12 VERY HARD

Two trains start from cities A and B toward each other at 60 km/h and 90 km/h. When they meet, the faster train has covered 180 km more than the slower. Find the distance between A and B.

Solution: Let time to meet = t. 90t – 60t = 180 → 30t = 180 → t = 6 h. Distance = (60+90)×6 = 150×6 = **900 km**

■ *Set up the difference-of-distances equation.*

Q13 VERY HARD

A man can row at 5 km/h in still water. If the stream speed is 1 km/h, it takes him 1 hour more to row upstream than downstream for the same distance. Find the distance.

Solution: Upstream speed = 4 km/h, Downstream = 6 km/h. Let dist = d. d/4 – d/6 = 1 → (3d–2d)/12 = 1 → d/12 = 1 → d = **12 km**

■ *Classic boat-and-stream variation: relates to relative speed.*

SECTION 5 — SHORTCUTS, TIPS & TRICKS

■ Speed Conversion Memory Trick

- km/h → m/s: Divide by 3.6 OR multiply by 5/18
- m/s → km/h: Multiply by 3.6 OR multiply by 18/5
- Trick: 5/18 ≈ 0.277 | 18/5 = 3.6
- Quick check: 18 km/h = 5 m/s (memorise this pair!)

■ Average Speed Shortcut

- Equal distances at speeds a & b: Avg = 2ab/(a+b) — NEVER (a+b)/2
- If speeds are in ratio 2:3, avg speed = 2×2×3/(2+3) = 12/5 units
- When three equal distances: Avg = 3abc/(ab+bc+ca)
- Equal TIME at speeds a & b: Avg = (a+b)/2 [Arithmetic mean — OK here!]

■ Train Problems — Quick Steps

- Step 1: Add lengths of objects being crossed
- Step 2: Find relative speed (add if opposite, subtract if same direction)
- Step 3: Time = Total Length / Relative Speed
- Convert km/h → m/s before dividing if lengths are in metres!

■ Speed-Time-Distance Triangle

- Draw a triangle: D on top, S and T on bottom
- Cover what you want: D=S×T | S=D/T | T=D/S
- If speed increases by x%, time decreases by x/(100+x)%
- If speed decreases by x%, time increases by x/(100-x)%

■ Percentage Change Tricks

- Speed 25% more → Time = 4/5 of original (i.e., 20% less)
- Speed 20% less → Time = 5/4 of original (i.e., 25% more)
- Speed doubled → Time halved
- Speed halved → Time doubled
- General: new time = original time × (original speed / new speed)

■ Late / Early Arrival Problems

- x minutes late at speed S■; y minutes early at speed S■
- Distance = S■×S■×(x+y) / (S■–S■) [S■ > S■]
- To be on time: use the time difference to find required speed
- Walking at 3/4 speed → time becomes 4/3 → extra time = T/3

■ Circular Track Shortcuts

- Same direction: first meet at Time = L / |S1-S2|
- Opposite direction: first meet at Time = L / (S1+S2)
- For A to gain one full round on B: L / |Sa-Sb|
- If A completes a lap in tA and B in tB, they meet at LCM(tA,tB)

■ Exam-Day Traps to Avoid

- TRAP 1: Not converting units — always check km/h vs m/s
- TRAP 2: Using arithmetic mean for average speed (equal distance case)
- TRAP 3: Forgetting to add lengths in train-crossing problems
- TRAP 4: Wrong relative speed direction (same vs opposite)
- TRAP 5: Confusing 'time taken' with 'extra time' in late/early problems

CKAT Exam Prep · Section A: Quantitative Aptitude · Topic 1: Time & Distance · Practice this topic until you can solve each problem in under 90 seconds.